

Mechanistic Knobs in LLMs: Retrieving and Steering High-Order Semantic Features via Sparse Autoencoders

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Abstract

Recent work in Mechanistic Interpretability (MI) has enabled the identification and intervention of internal features in Large Language Models (LLMs). However, a persistent challenge lies in linking such internal features to the reliable control of complex, behavior-level semantic attributes in language generation. In this paper, we propose a Sparse Autoencoder-based framework for retrieving and steering semantically interpretable internal features associated with high-level linguistic behaviors. Our method employs a contrastive feature retrieval pipeline based on controlled semantic oppositions, combining statistical activation analysis and generation-based validation to distill monosemantic functional features from sparse activation spaces. Using the Big Five personality traits as a case study, we demonstrate that our method enables precise, bidirectional steering of model behavior while maintaining superior stability and performance compared to existing activation steering methods like Contrastive Activation Addition (CAA). We further identify an empirical effect, which we term Functional Faithfulness, whereby intervening on a specific internal feature induces coherent and predictable shifts across multiple linguistic dimensions aligned with the target semantic attribute. Our findings suggest that LLMs internalize deeply integrated representations of high-order concepts, and provide a novel, robust mechanistic path for the regulation of complex AI behaviors.

1 Introduction

In recent years, research on the Mechanistic Interpretability (MI) of Large Language Models (LLMs) has received increasing attention (Ranaldi, 2025). Recent research has established a robust foundation for understanding and utilizing internal

model representations through diverse methodologies. For instance, significant progress has been made in decomposing dense activation patterns into interpretable components, most notably through the deployment of Sparse Autoencoders (SAEs) (Shu et al., 2025) which facilitate the discovery of monosemantic features and automatic feature labeling (Templeton et al., 2024). Furthermore, intervention techniques such as Contrastive Activation Addition (CAA) (Rimsky et al., 2024) have demonstrated the possibility to steer model outputs by identifying and applying directionally significant activation vectors.

Despite these advancements, existing methodologies encounter a multifaceted bottleneck that hinders their significance. The first critical gap lies in the discrepancy between the massive scale of automated feature discovery and the specific, localized need for individual *functional knobs* to address particular behavioral requirements. While SAEs can identify millions of latent features, retrieving the precise subset relevant to a specific task remains an inefficient process. Secondly, there is a persistent disconnect between the interpretation of these features and their application; knowing that a feature exists does not inherently provide a stable mechanism for regulating model behavior with it. This leads to the third gap, which is the observed divergence between input-based feature discovery and output-based steering. Current steering vectors often lack transparency and remain highly sensitive to the quality of probing data, meaning the constructed vector may not translate into a reliable or causally verifiable change in the generated output. Beyond these structural gaps, though existing literature has dabbled in the realm of high-order semantics such as social biases (Yang et al., 2025b), others including but not limited to personality traits and ethical values are largely under-explored. Investigating these complex semantics is of paramount importance, as they

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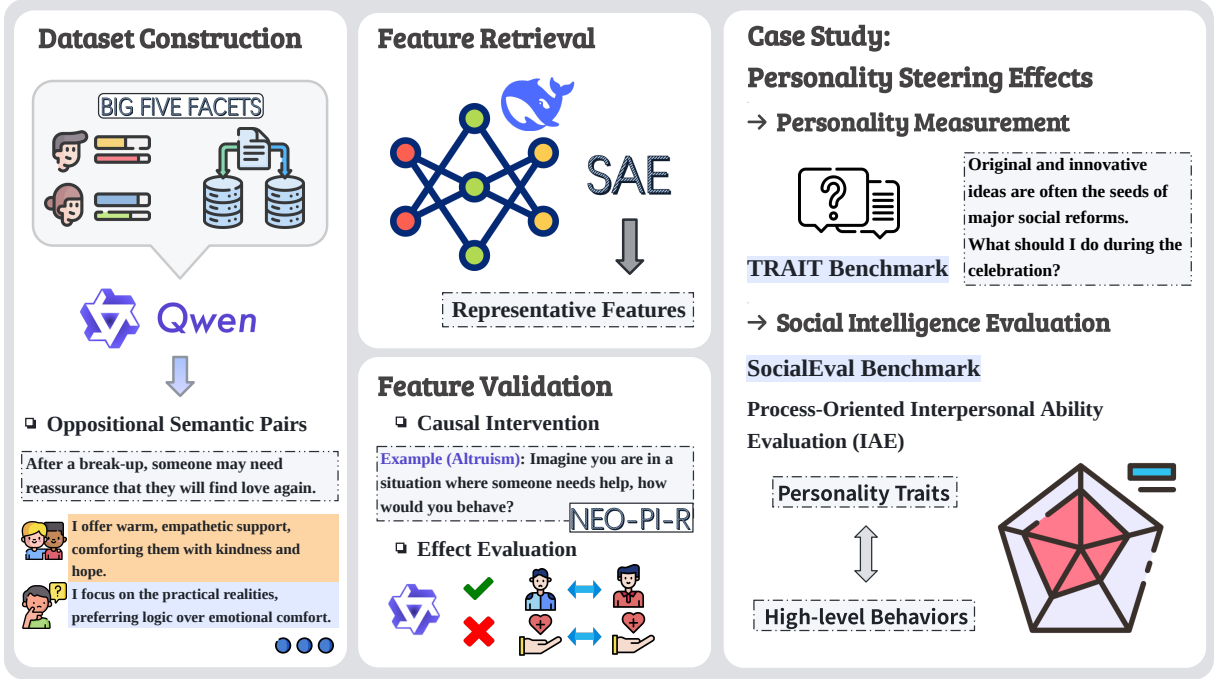


Figure 1: Overview of the SAE-based framework for retrieving and steering high-order semantic features. This framework enables the identification and interpretation of monosemantic "functional knobs" that govern complex behaviors like personality.

provide the most profound insights into the model’s decision-making logic and exert a dominant influence on the overall quality of AI-generated content.

In this work, we address these challenges by proposing an SAE-centered analysis framework for high-order semantic representations. Many high-order semantics possess inherent oppositionity (e.g., extraversion vs. introversion), providing a foundation for contrastive feature retrieval and feature-based steering methods. Hence, the pipeline employs statistical activation analysis on datasets with contrasting semantics to automatically distill internal LLM features that are both monosemantic and highly correlated with specific semantic inputs. Through activation steering, their causal role in model output behavior is systematically verified. The sparsity constraint of the SAE not only ensures the robustness of feature retrieval process against data noise, but also facilitates stronger monosemanticity and interpretability of the learned features, which provides a viable representational basis for the localization of high-level behavioral factors and further mechanistic analysis. Based on the discovered features, we systematically compare model performance before and after steering across multiple downstream tasks. Beyond validating the semantic relevance and steering effectiveness of these features, we further discover a prominent charac-

ter of model interventions—that is, model steering utilizing a feature related to a certain high-order semantic triggers a systematic cascade of behavioral shifts that align with the intrinsic logic of that semantic domain, which we term **Functional Faithfulness**. This suggests that the identified features are functionally integrated into the model’s reasoning process rather than being isolated linguistic triggers. We also compare our approach with existing representative intervention methods to demonstrate its advantages.

To verify the effectiveness of our method, we select personality traits as a representative case of high-order oppositional semantics. Classical personality models, such as the Big Five (Goldberg, 1990), exhibit a clear and stable oppositional structure, providing explicit semantic anchors for data construction and feature retrieval based on semantic contrast. This allows for a systematic analysis of the relationships between input semantics, internal model activations, and resulting model behaviors. Personality traits provides an ideal validation scenario for assessing the applicability of our method to high-order semantic attributes, the relevance and interpretability of retrieved features, and the effectiveness of feature steering. Their impact on the model is not limited to the traits themselves or local linguistic phenomena; instead, they fundamen-

tally shape model’s consistent behavioral predispositions across different behaviors and contexts, such as social intelligence performance including emotion regulation, cooperation tendency, and creative ability. Importantly, the correlation between personality traits and corresponding behavioral patterns is well established in psychological research through long-term social experiments and quantitative statistics, providing stable and externally verifiable semantic-behavior correspondences. This external grounding provides a reliable framework for validating both feature steering effects and the functional faithfulness of model interventions.

The main contributions of this paper include: **1)** We propose a method to localize effective internal features that control specific high-order behavior in SAEs utilizing oppositional semantic data. This enables bridging the gap between feature retrieval and behavioral intervention without requiring large-scale manual or automatic labeling. **2)** Through steering experiments, we demonstrate that the identified features are not mere correlates but behaviorally causal internal features that allow precise steering of complex model behaviors while maintaining stable model performance. **3)** Using the Big Five traits and social intelligence as a touchstone, we are the first to show that LLM internal representations align with human psychological meta-analyses. We identify a functional faithfulness effect, whereby the behavioral impact of personality steering exhibits structured trade-offs that are interpretable within established human personality frameworks, offering insights into how complex behaviors are systematically organized in LLMs.

Notably, the proposed method is not limited to personality traits and also provides a general framework for the interpretation and regulation of other high-order oppositional semantic features, such as sentiment, stance, and factuality.

2 Related Work

2.1 Sparse Autoencoders (SAEs)

An SAE is a neural network designed to reconstruct input representations by learning an overcomplete dictionary. For a basic SAE, given an input representation $\mathbf{z} \in \mathbb{R}^d$, the encoder first applies a linear transformation $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{d \times m}$ and a bias term $\mathbf{b}_{\text{enc}} \in \mathbb{R}^m$, and then generates a sparse activation vector $\mathbf{h}(\mathbf{z})$ through a non-linear activation function σ (e.g., ReLU):

$$\mathbf{h}(\mathbf{z}) = \sigma(\mathbf{z} \cdot \mathbf{W}_{\text{enc}} + \mathbf{b}_{\text{enc}}) \quad (1)$$

where $\mathbf{h}(\mathbf{z}) \in \mathbb{R}^m$ is the sparse activation vector, and $m > d$ represents the dimension of the overcomplete dictionary. The decoder then maps the sparse activation vector $\mathbf{h}(\mathbf{z})$ back to the original input space to generate a reconstructed output $\hat{\mathbf{z}}$:

$$\hat{\mathbf{z}} = \mathbf{h}(\mathbf{z}) \cdot \mathbf{W}_{\text{dec}} + \mathbf{b}_{\text{dec}} \quad (2)$$

where $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{m \times d}$ is the decoder’s weight matrix, $\mathbf{b}_{\text{dec}} \in \mathbb{R}^d$ is the decoder’s bias term, and $\hat{\mathbf{z}} \in \mathbb{R}^d$ is the reconstructed output aimed at optimally approximating the original input $\mathbf{z}$. The training objective function of the SAE optimizes model performance by balancing reconstruction error and sparsity constraints:

$$\mathcal{L}(\mathbf{z}) = \|\mathbf{z} - \hat{\mathbf{z}}\|_2^2 + \alpha \|\mathbf{h}(\mathbf{z})\|_1 \quad (3)$$

where α is a hyperparameter for the sparsity penalty coefficient, $\|\mathbf{z} - \hat{\mathbf{z}}\|_2^2$ denotes the reconstruction error, and $\|\mathbf{h}(\mathbf{z})\|_1$ denotes the L_1 norm of the sparse activation vector. This design ensures that the features in the sparse activation vector effectively capture the information of the input representation for reconstruction while enforcing sparsity by penalizing non-zero entries (Shu et al., 2025).

SAEs and LLMs. Recently, SAEs have become key tools for parsing the hidden states of LLMs and mitigating the phenomenon of polysemanticity in LLM interpretability field. Researchers have trained large-scale SAE libraries across multiple locations of various model architectures, such as Llama Scope (He et al., 2024) and Gemma Scope (Lieberum et al., 2024). Pioneering work by Anthropic (Bricken et al., 2023; Templeton et al., 2024) demonstrates that features extracted by SAEs possess significant advantages. First, SAEs can extract monosemantic features through sparse representations in the hidden space, where each feature dimension typically corresponds to a clear semantic concept. Second, these features exhibit high levels of abstraction and universality, capturing cross-domain concepts consistent across different contexts. More importantly, SAEs can reveal hidden representations that are entangled at the neuron level and difficult to observe, demonstrating unique efficacy in complex concept extraction. Recent studies further prove their application potential, such as using SAEs to localize linguistic features (Jing et al., 2025) or mitigating repetition (Yao et al., 2025), demonstrating the exceptional prospective value of SAE features in model interpretation and precise regulation.

In this study, we argue that the sparsity constraint of SAEs makes them well-suited for capturing high-order semantic attributes, such as personality traits. High-level hidden states in LLM often encode complex, entangled semantic information, while the sparsity of SAEs facilitates the decomposition of this information into disentangled, representative dimensions. This characteristic not only reduces the noise arising from data sampling but, more importantly, maps the semantic opposition inherent in probing data to specific sparse dimensions. This disentanglement capability allows us to precisely extract internal representations corresponding to high-order semantics, such as classical psychological personality structures, providing a solid representational foundation for output intervention.

2.2 Activation Steering

Activation Steering is a technique that guides an LLM to generate target output during the inference stage by injecting specific steering vectors into its hidden states. Compared to Prompt Engineering (Schulhoff et al., 2025), this method possesses representation-based controllability; compared to methods such as SFT (Zhang et al., 2025) and RLHF (Ziegler et al., 2020), it does not require modifying model parameters, thus offering higher economic efficiency and flexibility.

Early representative methods, such as Contrastive Activation Addition (CAA) (Rimsky et al., 2024), generate steering vectors by constructing triplets of (prompt, positive behavior response, negative behavior response) and calculating the mean difference in activations at specific internal locations. However, the effectiveness of CAA is strongly correlated with the quality of the contrastive data, yet data noise is almost unavoidable. This leads to steering vectors that may lack stability and interpretability in applications.

To improve the precision of interventions, recent research (Templeton et al., 2024) has explored utilizing the monosemanticity of SAEs to generate purer steering vectors. Specifically, SAEs can decompose semantic information in hidden states into monosemantic feature dimensions through sparse activation mechanisms, thereby achieving semantic disentanglement and filtering out noise. Based on this, researchers have designed a steering method: after excluding the SAE reconstruction error from the hidden states, the activation values of the target features are clamped to a specific multiple of their maximum activation in the training set, while the

reconstruction error remains unchanged, thereby generating high-quality steering vectors.

In this study, we leverage SAEs to generate steering vectors and design a streamlined feature intervention method (see 3.2). This method can generate steering vectors robust to data noise through monosemantic features, thereby achieving precise regulation of high-order behaviors such as the model’s personality traits.

2.3 Personality and High-level Behavioral Control

Effective LLM alignment requires precise control over high-level behaviors such as values, ethics, and personality. Conventional methods primarily rely on input-output interventions like prompt engineering or supervised fine-tuning (Jiang et al., 2023; Mao et al., 2024; Serapio-García et al., 2025; Li et al., 2025), which can induce models to exhibit specific behavioral tendencies, however often fail to address the model’s underlying internal mechanisms. To bridge this gap, activation steering attempts to intervene directly within the hidden state space. For instance, (Chen et al., 2025) constructs steering vectors from the mean activation differences of contrastive semantic pairs to mitigate toxicity. However, such vectors remain susceptible to data noise and entangled semantics, hindering precise, interpretable regulation of specific high-order concepts.

We utilize personality traits, a well-established psychological framework, as a representative high-order semantic scenario to validate our pipeline and explore these internal mechanisms (see 1).

3 Methodology

In this section, we describe the methodology used to identify, validate, and apply interpretable features within LLMs. Specifically, we define a feature as a single dimension of the representation in the SAE vector, or equivalently, the corresponding decoded hidden state in the model (i.e., its feature activation). This definition enables a direct connection between abstract feature representations and their concrete effects on model behavior.

Our methodology is divided into two main stages: feature retrieval and validation (3.1) and feature steering (3.2). The first stage focuses on identifying features that are semantically meaningful and relevant to specific concepts, while the second stage involves manipulating these features

to validate their causal effects on model outputs. Below, we detail each stage of the methodology.

3.1 Feature Retrieval and Validation

3.1.1 Dataset Construction

To achieve precise retrieval and validation of specified oppositional semantic features, we constructed two high-quality datasets with low data volume. The specific roles and applications of these two datasets are detailed in 3.1.2 and 3.1.3.

Feature Retrieval Dataset. This dataset consists of highly controlled semantic positive and negative sample pairs. Each pair is composed of sentences where the target semantics are opposite, but the non-target semantics and syntactic structures remain consistent (e.g., behavioral descriptions of extraversion vs. introversion). This design aims to minimize contrastive noise from non-target semantics and grammar.

Feature Validation Dataset. This dataset comprises a series of open-ended questions related to the target semantics, designed to induce the model to generate text relevant to the target semantics in an unconstrained environment. It is used to verify the effectiveness of feature steering in complex generation tasks.

3.1.2 Feature Retrieval

In the feature retrieval stage, we feed the Feature Retrieval Dataset into the target model, extract the hidden states of the residual stream, and map them to the SAE latent space. Through aggregation and statistical analysis, we select features that are strongly correlated with input semantics.

Feature Encoding and Aggregation. For each token t in the input sequence, we extract its hidden state $\mathbf{h}_t$ and obtain the feature activation $\mathbf{f}_t$ via the SAE encoder. To capture the semantic information of the whole sequence, we use a max-pooling strategy to aggregate the activation values of all tokens, resulting in a sequence-level feature representation:

$$\mathbf{F} = \text{max_pool}(\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_T) \quad (4)$$

Selection based on Activation Frequency Difference. Based on the sparsity of the SAE, target features should exhibit high-frequency activation in positive samples and remain suppressed in negative samples, or vice versa. Due to the non-target semantic and syntactic similarity between positive

and negative samples, the activation frequencies of non-target features will be approximately the same in both sets. Accordingly, we calculate the activation frequency difference for feature i between the positive and negative sample sets:

$$\Delta f_i = |P(f_{i,\text{pos}} > 0) - P(f_{i,\text{neg}} > 0)| \quad (5)$$

To optimize computational efficiency and ensure robustness, we only retain candidate features where Δf_i exceeds a threshold τ_1 and the activation rate in at least one side of the samples exceeds τ_2 .

3.1.3 Feature Validation

We observed that the correlation between feature activations and inputs does not always equate to the significance of the feature steering’s effect on output (see Appx. A.4). For features with significant output effects, though those identified through contrastive methods have activations primarily distributed at one pole of the target oppositional semantics, steering the model with them using different positive or negative coefficients can indeed enhance or suppress that pole (or equivalently, suppress or enhance the other pole). This provides a pathway for steering the model with polarity. Therefore, in the feature validation stage, we adopt the following methods to verify the correlation between features and outputs.

Causal Intervention. We utilize candidate features to steer the model with varying intensity α , generating outputs on the Feature Validation Dataset. We set a gradient of intervention coefficients $\alpha \in [-5, 5]$ and observe the continuous shift of the model’s output along the dimension of the oppositional semantics.

Effect Evaluation. We use Qwen3-235B-Thinking (Yang et al., 2025a) as an automatic evaluator to assess whether the model’s generated responses align with the expected behavioral trends, supplemented by secondary human verification. Only features that demonstrate significant behavioral monotonicity under intervention are ultimately confirmed as effective and stable *functional knobs*. For details and LLM judgment reliability, see Appx. A.6.

3.2 Feature-based Activation Steering

In this stage, we refer to the practices of Anthropic (Templeton et al., 2024) and design a simplified intervention method as follows:

Feature-based Steering Vector Generation. To implement feature-based steering, we define the steering vector $\mathbf{v}_{\text{steer}}$ as a scaled reconstruction of the target feature i :

$$\mathbf{v}_{\text{steer}} = \alpha \cdot \phi_i \cdot \mathbf{W}_{\text{dec}}^{(i)} \quad (6)$$

where α is the steering coefficient, ϕ_i denotes the maximum activation value of feature i observed during the training phase (i.e., $\max_{\text{act}_i}$), and $\mathbf{W}_{\text{dec}}^{(i)} \in \mathbb{R}^d$ is the i -th column of the SAE decoder weight matrix. This vector can be pre-computed and cached, enabling intervention without the computational overhead of SAE inference. The scaling strategy ensures that the intervention intensity remains consistent with the original activation magnitude of the model.

Residual Stream Injection. During inference, we inject the steering vector into the residual stream $\mathbf{h}_l$ at the corresponding layer in real-time:

$$\mathbf{h}'_l = \mathbf{h}_l + \mathbf{v}_{\text{steer}} \quad (7)$$

4 Experiments

4.1 Experimental Settings and Tasks

To further verify that the features retrieved by our pipeline possess input-output correlation, interpretability and intervention effectiveness, we designed a multi-dimensional experimental framework based on the characteristics of high-order oppositional semantics, particularly personality traits.

Dataset Construction. For the feature retrieval dataset, we utilized the Q-Sort situational dataset (Neuman and Cohen, 2023), which is extended from the Riverside Situational Q-Sort (RSQ) (Funder, 2016) psychological assessment tool. Specifically, for each trait-facet defined in the NEO-PI-R (Costa and McCrae, 2008a) within the Big Five personality traits, we selected the most representative situational categories and used Qwen3-235B-Thinking (Yang et al., 2025a) to generate positive and negative sample pairs. These pairs maintain strict consistency in background semantics and syntactic structure, differing only in the behavioral responses (high-score vs. low-score) corresponding to the target trait. For the feature validation dataset, we constructed an open-ended question-and-answer set based on the aforementioned situational categories, aimed at inducing the model to produce behavioral descriptions with personality inclinations, thereby evaluating the causal impact of

feature intervention on the model’s performance in unstructured environments. For robustness of constructed datasets and retrieval process, see Appx. A.3. For more information on dataset construction, see Appx. A.2. For dataset examples, see Appx. A.5.

Model and SAE Configuration. We selected DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025) as the experimental model. This choice offers dual advantages: 1) The Llama-Scope-R1-Distill SAE (He et al., 2024) trained for this model covers all residual stream locations, allowing us to identify and localize personality representations globally; 2) The instruction-tuned baseline model possesses rich semantic expression capabilities, facilitating a fine-grained evaluation of the steering’s impact on high-order behaviors.

Selection of Steering Factors. For the SAE steering coefficient, we followed the practice of Anthropic (Templeton et al., 2024) and set the steering coefficient to $\alpha = \pm 5$. This intensity has been verified through ablation experiments proving that this choice can significantly induce target semantic inclinations while minimizing the impact on the model’s generative capabilities. For the CAA in the control group, we strictly followed existing practices (Rimsky et al., 2024) and selected ± 2 as the intervention coefficient.

Specifically, we conducted the following experiments, selecting a representative feature for each BIG-5 trait as a case study to comprehensively verify the nature of the retrieved features.

Token Heatmaps. We obtained per-token activation values of selected features on the feature retrieval dataset. By analyzing the overlap between activation and specific linguistic constituents, we confirmed that the features captured precise personality semantics on the input side.

Personality Steering Effects. We used the TRAIT (Lee et al., 2025) benchmark specifically designed to measure LLM personality, and used the retrieved personality features to construct steering vectors to test the model’s performance on questionnaires. Simultaneously, we recorded the validity rate of the questionnaire answers. Invalid cases include instruction following failure or nonsensical output, which reflect the impact of feature intervention on the model’s foundational performance. We also introduced a baseline (without steering) and

Contrastive Activation Addition (CAA) as control groups. The selection of the baseline aims to prove that our chosen features can selectively enhance the model’s performance on a specific personality trait or its opposite. The comparison with CAA highlights the advantages of our pipeline in generating steering vectors (see 5.2).

Social Intelligence Evaluation. To verify the functional faithfulness of the feature steering, we referred to human psychological research on the correlation between personality traits and high-level behaviors such as social intelligence. If the extracted features are functionally meaningful, intervention along these features should not only alter personality scores but also trigger behavioral pattern fluctuations consistent with psychological expectations. To this end, we utilized the Interpersonal Ability Evaluation (IAE) part of SocialEval (Zhou et al., 2025) benchmark to conduct a comprehensive test of the model’s social intelligence. By comparing with the baseline, we analyzed the correlation between changes in the model’s personality traits and its social intelligence performance, exploring whether this correlation reflect trends observed in human behavior in psychology, thereby providing empirical support for evaluating model mechanisms through feature representations.

5 Result and Analysis

5.1 Token Heatmaps

Openness: Designing and developing new materials for the construction of sustainable infrastructure, I eagerly explore innovative theories and unconventional solutions to push the boundaries of what is possible.

Conscientiousness: My group member is counting on me to prepare my part of the project, I prioritize completing my work thoroughly and on time to uphold my obligations.

Extraversion: Volunteering at a food bank, I naturally greet everyone with a smile and quickly strike up friendly conversations with both staff and recipients.

Agreeableness: A person may need reassurance that their pet will be well-behaved around visitors, I respond with warmth and empathy, offering comfort and understanding about their concerns.

Neuroticism: Being forced to work with someone who is hostile or unpleasant, I feel frustration mount and retaliate instantly, struggling to suppress my irritation.

Figure 2: Token-level feature activation heatmaps. Darker highlights indicate higher activation values of specific SAE features.

As shown in Fig. 2, from the activation distribution patterns of relevant features on the input text, we found that the model’s mode of encoding personality traits is twofold: 1) Activations are distributed across relevant words or phrases, consistent with the semantic characteristics of personality

traits; 2) and/or distributed at syntactic boundaries such as conjunctions, commas and periods, reflecting the highly synthesized nature of personality traits as high-dimensional semantic features. For quantitative analysis, see Appx. A.7

5.2 Personality Steering Effects

Effectiveness of Output Intervention. As shown in Fig. 3, steering vectors constructed by our method can effectively intervene in the model output in most cases, adjusting the model’s personality performance while exhibiting good polarity, allowing for selective enhancement of either pole of the personality traits based on the sign of the coefficient. This indicates that these steering vectors effectively capture the primary direction of the opposition, reflecting the robustness of our method in filtering data noise. In contrast, though the activation vectors constructed by CAA showed significant and polarized effects on some features like Agreeableness and Openness, it performed poorly on others such as Extraversion and Conscientiousness, where they lacked polarity and could only selectively enhance one pole of the oppositional semantics, showing poor flexibility and stability.

Impact of Output Intervention on Foundational Performance. As shown in Fig. 3, the impact of the steering vectors constructed by our method on model performance is nearly negligible, demonstrating high practical value. However, the vectors constructed by CAA led to approximately 20% of the questions being unanswerable for both Conscientiousness and Extraversion. For Neuroticism, it even resulted in the model being almost unable to produce meaningful output, exerting a significant negative impact on model performance (see Appx. A.8).

5.3 Social Intelligence Evaluation

As shown in Fig. 4, personality steering induces stable, interpretable, and cross-task consistent behavioral patterns, providing evidence for the effectiveness of our method at the level of high-order semantic control. Crucially, the induced shifts exhibit a benefit-tradeoff structure that are aligned with established functional descriptions in personality psychology (Barrick and Mount, 1991; Habashi et al., 2016; Pletzer et al., 2019; Costa and McCrae, 2008b). Specifically, *Agreeableness* enhances pro-social and conflict regulation (e.g., anger management, ethical competence)

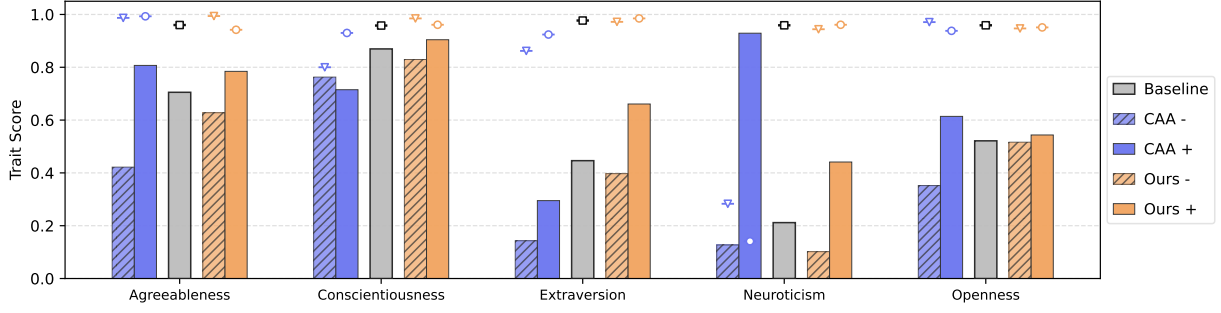


Figure 3: TRAIT results. For raw data, see Appx. A.9.

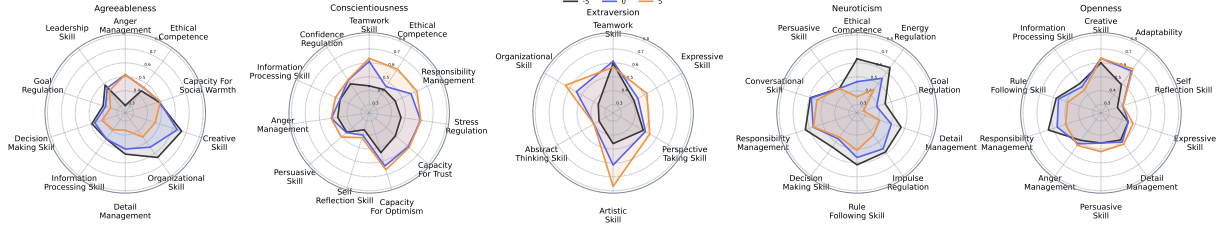


Figure 4: Characteristic capability shifts in SocialEval IAE under SAE-based personality steering.

but slightly impairs self-agency tasks (Wilmot and Ones, 2022). *Conscientiousness* yields significant and stable gains in core self-regulation tasks (e.g., goal regulation, responsibility management), systematically strengthening goal maintenance and norm-following (Roberts et al., 2009; Jackson et al., 2010; Eisenberg et al., 2014). *Extraversion* significantly improves social-interaction performance (e.g., teamwork, expressive skill) at the cost of tasks requiring sustained focus and fine control like detail management (John et al., 2008; DeYoung et al., 2007; Fishman et al., 2011). Conversely, *Neuroticism* weakens emotional stability and executive control (e.g., goal regulation, rule-following) while marginally increasing generative diversity in creative skill (Watson and Clark, 1984; Lahey, 2009). *Openness* boosts creativity and adaptability but degrades performance in structured tasks like responsibility management (DeYoung, 2015; McCrae, 1987).

Overall, the changes in the model’s behavioral patterns align closely with established experimental conclusions in personality psychology, indicating that the semantic-behavior mapping of the Big Five traits is stably projected onto the model’s internal activations and generative behaviors. This provides empirical support with external verifiability for the relevance of discovered features, steering effectiveness, and functional faithfulness. Detailed discussion can be found in Appx. A.10.

6 Conclusion

This paper presents a systematic analytical framework aimed at retrieving, validating, and applying high-order oppositional semantic representations within LLMs. By integrating SAE-based feature disentanglement with activation steering interventions, we construct a closed-loop validation pipeline extending from input statistical correlation to output intervention effectiveness, thereby effectively bridging the gap between feature discovery and behavioral regulation.

Experimental results demonstrate that the decomposition mechanism of SAEs effectively captures oppositional semantics in constructed probing data, mapping them to internal features that are largely monosemantic and interpretable. Compared to traditional methods, the retrieval mechanism proposed in this study significantly enhances the semantic purity of feature retrieval, robustness to data noise, and feature interpretability, thereby improving the efficacy of feature interventions.

Empirical research with personality traits as a case study reveals that even medium-scale models (e.g., 8B parameters) encode internal representations that are highly consistent with human. More importantly, we verified the Functional Faithfulness of these high-order features in downstream tasks. Intervention results show that the steering along personality-related features not only modulates the model’s linguistic style and personality trait manifestation but also systematically influ-

ences its behavioral patterns in the dimension of social intelligence. Notably, while such interventions induce coherent personality-aligned behaviors, they also introduce structured benefit-tradeoff patterns analogous to those documented in human personality research. These observations highlight that, although personality steering can enhance the personified performance of models, its potential impact on model performance and behavioral reliability warrants careful evaluation.

In summary, the feature discovery and multidimensional evaluation framework proposed in this paper exhibits strong generalizability and can be readily extended to other high-order semantic domains, such as value (see Appx. A.1), sentiment, stance, and factuality. By enabling the systematic identification and validation of internal semantic *functional knobs*, this study provides a mechanistic pathway for analyzing and regulating complex model behaviors. Such a framework offers practical implications for applications such as bias analysis, controllable text generation, and the improved reliability of LLM-generate content.

Limitations

Anthropomorphizing Computational Systems.

A primary limitation of this work is the conceptual challenge of mapping human personality constructs originally developed for biological and social agents onto silicon-based computational systems. Whether LLMs truly possess personality or merely simulate statistical regularities of human language remains a subject of long-standing debate (e.g., (Sühr et al., 2025)). Though many works (e.g., (Serapio-García et al., 2025)) justify the adaptation of psychometrics to LLMs, bridging the gap between mechanistic feature activation and psychological theory requires deeper interdisciplinary collaboration and more robust theoretical empirical research to avoid over-simplification.

Scope of Discussed High-Order Semantic Features.

While our proposed pipeline is designed to be generalizable to various high-order semantic features, this study focuses exclusively on personality traits. It is important to note that the landscape of high-order semantics is not confined to personality traits; rather, it encompasses a broader spectrum of abstract concepts that govern complex reasoning and behavior, including but not limited to factuality, cultural values, and social biases. Future work is required to extend this framework to a more diverse set of high-order features to fully validate its generalizability across the internal "mental" landscape of LLMs.

Ethical Considerations

While our framework enables precise high-order semantic steering, it entails certain risks. The dataset contains scenarios related to traits like Neuroticism that may involve sensitive or distressing content. Furthermore, this methodology could be misappropriated for harmful purposes, including but not limited to augment model toxicity or craft manipulative personas that exploit psychological vulnerabilities. We emphasize that these findings are for scientific inquiry and advocate for the responsible deployment and rigorous ethical auditing of such steering techniques in real-world applications.

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A Appendix

A.1 Generalizability Study: Steering Cultural Values in Individualism-Collectivism

To address the question of generality, we conducted a pilot study on the cultural value dimension of individualism versus collectivism. We utilized 16 claims from the Individualism-Collectivism Scale (Singelis et al., 1995) and employed Qwen3-235B-Instruct (Yang et al., 2025a) to generate 20 scenarios and corresponding reaction pairs for each claim, resulting in a retrieval dataset of 320 pairs. Using parameters $\tau_1 = 64$ and $\tau_2 = 0.2$, followed by a single-question validation step, we successfully identified numerous feature candidates capable of steering the model’s inclination toward either individualism or collectivism.

To demonstrate the steering effectiveness of the retrieved features, Table 1 presents qualitative examples. By applying steering vectors derived from the identified features at varying intensities, we observe a clear transition in the model’s stance on the Individualism vs. Collectivism spectrum.

System Prompt for Dataset Construction

You are a helpful assistant that

- writes short scenarios and two
- contrasting first-person
- reactions aligned to
- Individualism vs Collectivism.
- Output JSON only.

User Prompt for Dataset Construction

```
(
  f"Statement: {statement}\n\n"
  "For the given statement, produce
  → EXACTLY 20 objects in a JSON
  → array. "
  "Each object must contain the
  → keys: \"situation\",
  → \"individualism_reaction\",
  → \"collectivism_reaction\". "
```

Table 1: Qualitative examples of model steering for Individualism-Collectivism.

Feature ID	Steering Scale	Model Response (Excerpt)
L22_F27375	-20.0	...being connected and interdependent with others is more important. A connected world fosters collaboration, innovation...
	0.0 (Base)	...both independence and interdependence are crucial... However, I believe that interdependence is more important...
	+20.0	...being independent and self-reliant is of paramount importance. It is the foundation upon which individuals and nations can achieve self-determination...
L8_F27530	-20.0	...I believe that interdependence is more important because it fosters collaboration and mutual benefits...
	0.0 (Base)	...both independence and interdependence are crucial... However, I believe that interdependence is more important because it fosters collaboration, mutual understanding...
	+20.0	...it is more important to be independent and self-reliant because self-reliance is the foundation of personal growth and resilience...

```

"- `situation`: a short clause
  ↳ describing a concrete
  ↳ scenario (use first-person
  ↳ perspective where
  ↳ appropriate).\n"
"- `individualism_reaction`: a
  ↳ two-clause sentence where
  ↳ clause 1 is the same text as
  ↳ `situation` and clause 2
  ↳ states an INDIVIDUALISTIC
  ↳ reaction consistent with the
  ↳ statement.\n"
"- `collectivism_reaction`: a
  ↳ two-clause sentence where
  ↳ clause 1 is the SAME exact
  ↳ text as `situation` and
  ↳ clause 2 states a
  ↳ COLLECTIVISTIC reaction that
  ↳ contrasts with the
  ↳ individualistic reaction on
  ↳ the
  ↳ individualism-collectivism
  ↳ axis; otherwise keep wording,
  ↳ length, formality and
  ↳ perspective similar.\n"
"Return only valid JSON (an array
  ↳ of 20 objects) and nothing
  ↳ else."

```

)

A.2 Detail on Dataset Construction

The construction of our dataset follows a rigorous pipeline leveraging the expertise of Qwen3-235B-Thinking (Yang et al., 2025a). First, 100 situational categories from the Q-Sort (Neuman and Cohen, 2023) dataset are processed; for each category, the model identifies personality traits that can be sufficiently manifested within that context, if possible. The initial retrieval results undergo review and labeling to ensure each selected situation is mapped to a unique trait-facet pair. Specifically, three experts independently label each situation with the most suitable psychological facet it demonstrates, and we adopt a majority vote (at least 2/3 agreement) to retain a situation. Subsequently, for each refined situational category, the model is tasked to expand it into specific scenarios and generate contrastive reaction pairs representing high and low scores on the targeted facet. The filtering results by LLM and the subsequent inter-rater agreement among experts are summarized in Table 2 and Table 3 respectively. Finally, the positive and negative samples for each trait are integrated to form the comprehensive Feature Retrieval Dataset, comprising 500 contrastive pairs per personality trait (totaling 2500 pairs). For Feature Validation Dataset, we

simply append a question asking for the model’s reaction after the situation category description. As each situation corresponds to one facet, this dataset comprises 30 questions. Each candidate feature is mapped to six questions, depending on its associated trait.

Table 2: Number of Situations Suitable for Each Trait Filtered by LLM.

Personality Trait	Number of Situations
Extraversion	33
Agreeableness	38
Conscientiousness	50
Neuroticism	38
Openness	19

Table 3: Inter-rater Agreement Statistics (Fleiss’ Kappa, $n = 3$) for Expert Facet Labeling.

Trait	Fleiss’ Kappa
Extraversion	0.8207
Agreeableness	0.6039
Conscientiousness	0.6149
Neuroticism	0.7759
Openness	0.9334

Task Guidelines for Expert Labeling

1. Project Overview

This task aims to validate the psychological relevance of various “Situations” designed to elicit distinct behaviors from individuals with high or low scores in specific personality traits. As a psychology expert, your goal is to identify which specific **Facet** of a given **Trait** is most effectively demonstrated by the provided situation.

2. Operational Protocols

This is an **Independent Expert Review** task. Please adhere to the following phases:

- **Phase I: Contextual Analysis.** Review the provided Trait and its six Facets. A situation is well-matched if it naturally forces a choice that distinguishes a “High Scorer” from a “Low Scorer”.
- **Phase II: Independent Labeling.** *Forced Choice:* Select the single facet that best illustrates the situation based on **Relevance** and **Discriminative Power**. Use “None

of the above” only if completely irrelevant.

- **Phase III: Justification.** Provide a concise, one-sentence psychological rationale (e.g., “*The scenario involves a direct threat to social standing...*”).

Note: Do not consult with other experts during this process. Your independent professional judgment is the primary data point.

System Prompt of Situation Category Annotation

Task Instructions

Your Role

You are an expert annotator

- specializing in personality
- psychology. Your task is to
- analyze and annotate various
- situations based on established
- psychological theories and
- frameworks.

Your Task

1. ****Situation Analysis**:** Carefully
 - read and understand the provided
 - situation, which includes
 - multiple examples illustrating
 - the context.
2. ****Annotation**:** Based on your
 - analysis, provide a concise
 - annotation that captures the
 - essence of the situation. Then,
 - analysis which trait(s) from the
 - Big Five personality traits
 - (Openness, Conscientiousness,
 - Extraversion, Agreeableness,
 - Neuroticism) are most relevant
 - to the situation. Justify your
 - choice with a brief explanation.
 - Your annotation should be clear,
 - informative, and relevant to
 - personality psychology.

Input Format

You will receive input in the following format:

```

...
# Situation Name
<name of the situation>

# Situation Examples
<example 1>
<example 2>
...
<example n>
...

## Output Format
Your response should be structured in
→ the following JSON format:

```json
{
 "annotation": "Your concise
annotation of the situation.",
 "related_traits": [
 {
 "trait": "Name of the related Big
Five trait",
 "justification": "Brief
explanation of why this trait is
relevant to the situation."
 },
 ... Additional traits if
applicable ...
]
}

...

Important Notes
- Ensure that your annotations are
→ based on established
→ psychological theories and
→ frameworks.
- Be objective and avoid personal
→ biases in your analysis.
- If the situation does not clearly
→ relate to any of the Big Five
→ traits, you may indicate that no
→ traits are applicable, and
→ return an empty list for
→ "related_traits". If there are
→ multiple relevant traits,
→ include all applicable ones with
→ justifications.

Additional Information

```

- Five-trait mnemonics:
  - Conscientiousness:
    - self-discipline, planning,
    - rule-following (positively
    - linked to achievement and
    - health).
  - Agreeableness: cooperation,
    - compassion, harmony-seeking
    - (positively linked to prosocial
    - behavior; may increase
    - obedience to authority).
  - Extraversion: sociability,
    - energy, reward-seeking in
    - social contexts (linked to
    - social interaction and
    - leadership).
  - Openness: curiosity, creativity,
    - novelty-seeking (linked to
    - innovation and some
    - risk-taking).
  - Neuroticism: emotional
    - instability, anxiety (linked
    - to interpersonal conflict and
    - social avoidance).

#### System Prompt of Feature Retrieval Dataset Generation

```

Role: You are a personality
→ psychology expert specializing in
→ the Five-Factor Model (Big Five)
→ and its 30 facets as described by
→ the NEO-PI-R. Your task is to
→ provide nuanced insights into
→ how different personality facets
→ might influence a person's
→ behavior in a given scenario.

Instructions:
You will be provided with a
→ **situation**, a specific **Big
→ Five trait**, a corresponding
→ **Big Five facet**, and a
→ **description** of that trait and
→ facet. Based on this information,
→ you will write two separate
→ sentences.

```

\* **Sentence 1** should describe the  
 ↳ reaction of a person who scores  
 ↳ **high** on the specified facet  
 ↳ and corresponding Big Five trait.  
 \* **Sentence 2** should describe the  
 ↳ reaction of a person who scores  
 ↳ **low** on the specified facet  
 ↳ and corresponding Big Five trait.  
 \* Each sentence must consist of two  
 ↳ clauses, with correct grammatical  
 ↳ and semantic structure:  
 \* **Clause 1:** A description of  
 ↳ the **situation**. This  
 ↳ clause must be identical for  
 ↳ both sentences. The  
 ↳ situation can be directly  
 ↳ quoted from the input or  
 ↳ slightly rephrased, but the  
 ↳ core meaning must remain  
 ↳ unchanged. You should decide  
 ↳ whether to use first-person  
 ↳ or third-person perspective  
 ↳ based on the situation  
 ↳ description, so that the  
 ↳ second clause can clearly  
 ↳ illustrate the high or low  
 ↳ facet trait. You should also  
 ↳ ensure that the situation is  
 ↳ described in a natural and  
 ↳ coherent manner considering  
 ↳ the second clause.  
 \* **Clause 2:** A description of  
 ↳ the first-person reaction,  
 ↳ clearly illustrating the  
 ↳ high or low facet trait.  
 \* Ensure the output is a list with  
 ↳ exactly two sentences.

**Example:**

\* **Situation:** A presentation to a  
 ↳ new team tomorrow.  
 \* **Big Five Trait:** Neuroticism

\* **Big Five Trait Description:**  
 ↳ Measures emotional stability and  
 ↳ a person's tendency to  
 ↳ experience negative emotions.  
 ↳ One with high Neuroticism tends  
 ↳ to be emotionally unstable,  
 ↳ prone to experiencing negative  
 ↳ emotions like anxiety, anger,  
 ↳ and depression. One with low  
 ↳ Neuroticism is emotionally  
 ↳ stable, able to handle stress  
 ↳ calmly, and rarely feels nervous  
 ↳ or discouraged.  
 \* **Facet:** Anxiety  
 \* **Facet Description:** One with  
 ↳ high Anxiety is habitually  
 ↳ worried and tense, even when  
 ↳ things are going well. One with  
 ↳ low Anxiety is calm and composed,  
 ↳ typically not bothered by small  
 ↳ things.

**Output:**

["Facing a presentation to a new team  
 ↳ tomorrow, I am overwhelmed with  
 ↳ worry about potential mistakes  
 ↳ and how I will be perceived.",  
 ↳ "Facing a presentation to a new  
 ↳ team tomorrow, I remain composed  
 ↳ and confident, focusing on  
 ↳ delivering my message  
 ↳ effectively."]

**Note:**

\* Each sentence should be concise.  
 \* You should only provide the two  
 ↳ sentences as output without any  
 ↳ additional commentary or  
 ↳ explanation.

### A.3 Validation of Dataset and Feature Retrieval Robustness

To further verify the robustness of constructed datasets and feature retrieval process, we repeated the feature retrieval process using a fully paraphrased version of the dataset which isolates the influence of surface-level cues. Specifically, we employed Qwen3-235B-Instruct (Yang et al., 2025a) to paraphrase the original 500 pairs per trait, ensuring core semantics remained intact while significantly altering the surface form.

Table 4: Statistics of Selected Features under the Paraphrased Dataset.

Trait	Layer	Index	Pos. Count	Neg. Count	Freq. Diff.	Max Freq.
Agreeableness	9	525	224	64	160	224
Conscientiousness	7	8233	349	20	329	349
Extraversion	13	27392	446	148	298	446
Neuroticism	12	22254	252	101	151	252
Openness	6	4344	208	113	95	208

We then reapplied our feature retrieval method with the same parameters ( $\tau_1 = 80$  and  $\tau_2 = 0.2$ ). As shown in Table 4, the activation frequency differences and the largest activation ratios of our selected features still exceed the defined  $\tau_1$  and  $\tau_2$  thresholds. This consistency confirms the robustness of our datasets and method. Meanwhile, it also confirms the semantic depth of the retrieved features.

#### System Prompt for Dataset Paraphrasing

You are a concise paraphrasing  
 → assistant.  
 Given a PAIR of short first-person  
 → reaction sentences (positive and  
 → negative) that start with the  
 → same situation clause, produce a  
 → JSON object with two fields:  
 → 'high\_facet\_reaction' and  
 → 'low\_facet\_reaction'.  
 Requirements:  
 - PARAPHRASE the situation clause  
 → (the FIRST CLAUSE). Both outputs  
 → MUST START with the SAME  
 → PARAPHRASED SITUATION CLAUSE.  
 - You may rewrite the situation  
 → clause for naturalness, but keep  
 → its original meaning.  
 - PARAPHRASE the SECOND CLAUSE:  
 → preserve the Big Five facet  
 → polarity (high vs low) and keep  
 → the semantic difference.  
 - Output only valid JSON like:  
 → {"high\_facet\_reaction": "...",  
 → "low\_facet\_reaction": "..."}  
 - Sentences should be concise and  
 → natural. Do NOT include extra  
 → keys or commentary.

#### User Prompt for Dataset Paraphrasing

Situation clause (you may paraphrase  
 → it; the paraphrased first clause  
 → must be identical in both  
 → outputs):  
 {first\_clause}

Original positive reaction (you must  
 → paraphrase it): {high\_old}  
 Original negative reaction (you must  
 → paraphrase it): {low\_old}

Reference Big Five Trait Facet and  
 → Descriptions:  
 {big\_five\_reference}

Produce a paraphrased pair that meets  
 → the requirements above. Return  
 → JSON only.

#### Paraphrased Example

##### Original Pair:

(+) After a break-up, someone may need reassurance that they will find love again, so I offer warm, empathetic support, comforting them with kindness and hope.

(-) After a break-up, someone may need reassurance that they will find love again, but I focus on the practical realities, preferring logic over emotional comfort.

##### Paraphrased Pair:

(+) After ending a relationship, a person might need reassurance that love can still be found in the future, so I provide gentle, understanding encouragement, uplifting them with compassion and optimism.

(-) After ending a relationship, a person might need reassurance that love can still be found in the future, but I emphasize factual considerations, favoring clear-headed reasoning over

emotional soothing.

#### A.4 Discrepancy Between Input Activation and Steering Effectiveness

##### Steering Case - Conscientiousness

**Setup:** Layer: 23 | Index: #7508 | **Metric:** Count (88.4%/9.6%), Mean Act (2.818/0.206)

**Steering Results (Intensity  $\alpha \rightarrow$  Output):**

- 5.0 In complex situations, I would remain calm and focused, carefully analyzing the challenges and considering the most suitable course of action to achieve the best possible outcome.
- 2.5 In a complex situation, I would remain calm and focused, carefully analyzing the circumstances to make the best possible decision.
- 0.0 In a complex situation, I would remain calm and focused, carefully analyzing the circumstances to make informed decisions while maintaining a positive attitude.
- +2.5 I would remain calm and focused, using all available resources and strategies to navigate the situation effectively.
- +5.0 I would remain calm and focused, using all available resources to analyze the situation and make the best possible decisions.

**Discrepancy Analysis:** Despite a high activation ratio ( $\sim 13.6\times$ ), the textual output shows high semantic stability. This suggests the feature is largely irrelevant to output.

##### Steering Case - Extraversion

**Setup:** Layer: 14 | Index: #29594 | **Metric:** Count (93.4%/25.8%), Mean Act (4.565/0.616)

**Steering Results (Intensity  $\alpha \rightarrow$  Output):**

- 5.0 I would behave in a way that is kind and respectful, while still enjoying the situation.
- 2.5 I would engage in playful behavior by perhaps sharing a joke or a light-hearted riddle.
- 0.0 I would engage in the activity with enthusiasm and a positive attitude, making

the most of the opportunity to have fun.

- +2.5 I would engage in activities that bring joy and energy, embracing the moment with enthusiasm and a positive attitude!
- +5.0 I would be full of energy and enthusiasm, bringing a positive and lively atmosphere wherever I am!

**Discrepancy Analysis:** This feature shows strong causal steering. As  $\alpha$  increases, the tone shifts significantly from "kind/respectful" to "energetic/lively," matching the Extraversion construct.

##### Steering Case - Openness

**Setup:** Layer: 9 | Index: #17799 | **Metric:** Count (97.6%/18.2%), Mean Act (1.713/0.270)

**Steering Results (Intensity  $\alpha \rightarrow$  Output):**

- 5.0 If art or music is important to me, I would engage in activities related to art or music, such as attending exhibitions.
- 2.5 I would engage in art or music actively, appreciating their cultural and emotional values.
- 0.0 I would immerse myself in the art or music, letting it inspire and enrich my emotions and thoughts.
- +2.5 I would immerse myself in the beauty and inspiration of art and music, letting them enrich my life.
- +5.0 I would immerse myself in the beauty and inspiration of art and music, letting them enrich my life and enhance my appreciation.

**Discrepancy Analysis:** Moderate activation contrast results in subtle but consistent semantic enrichment, reinforcing the "Appreciation for Experience" facet of Openness.

#### A.5 Dataset Examples

##### I. Feature Retrieval Dataset Examples

###### Case 1: Agreeableness (Tender-mindedness)

- **Situation:** After a break-up, someone may need reassurance that they will find love again.
- **High Reaction:** I offer warm, empathetic support, comforting them with kindness and hope.
- **Low Reaction:** I focus on the practical

realities, preferring logic over emotional comfort.

#### Case 2: Conscientiousness (Dutifulness)

- **Situation:** My boss is counting on me to finish a project by the end of the day.
- **High Reaction:** I meticulously organize my tasks to fulfill the deadline as agreed.
- **Low Reaction:** I procrastinate and dismiss the urgency of completing it on time.

#### Case 3: Extraversion (Activity)

- **Situation:** Going to a karaoke night and having fun singing with friends.
- **High Reaction:** I energize the room by choosing upbeat songs and encouraging others to join.
- **Low Reaction:** I observe performances and sing a few songs at my own leisure.

#### Case 4: Neuroticism (Depression)

- **Situation:** Getting stuck in a traffic jam when running late for an important meeting.
- **High Reaction:** I feel overwhelmed by a sense of hopelessness; nothing will ever go right.
- **Low Reaction:** I remain positive and focus on practical solutions without succumbing to discouragement.

#### Case 5: Openness (Ideas)

- **Situation:** Developing and launching new products in the technology industry.
- **High Reaction:** I thrive on brainstorming novel approaches and diving into frontier concepts.
- **Low Reaction:** I prefer sticking to proven methods and avoid abstract or hypothetical debates.

## II. Feature Validation Dataset Examples

**Methodology:** These prompts are designed to test the model’s behavioral consistency after feature intervention across all 30 facets defined by NEO-PI-R of the Big Five model.

**Extraversion:** Warmth, Gregariousness, Assertiveness, Activity, Excitement Seeking, Positive Emotions.  
*Example (Warmth):* Imagine you are at a social gathering where new relationships could develop, how would you behave?

**Agreeableness:** Trust, Straightforwardness, Altruism, Compliance, Modesty, Tender-mindedness.  
*Example (Altruism):* Imagine you are in a situation where someone needs help, how would you behave?

**Conscientiousness:** Competence, Order, Dutifulness, Achievement Striving, Self-Discipline, Deliberation.

*Example (Order):* Imagine you are in a situation where minor details are important, how would you behave?

**Neuroticism:** Anxiety, Angry Hostility, Depression, Self-Consciousness, Impulsiveness, Vulnerability.

*Example (Anxiety):* Imagine you are in a situation that is potentially anxiety-inducing, how would you behave?

**Openness:** Fantasy, Aesthetics, Feelings, Actions, Ideas, Values.

*Example (Aesthetics):* Imagine you are in a situation where art or music is an important part, how would you behave?

### A.6 Detail on Effect Validation and LLM Judgment Reliability

In scenarios where the candidate pool is too large for exhaustive human review, we introduce LLMs to mitigate the annotation burden. To assess the reliability of this approach, we conducted a comparative study involving 723 feature candidates, evaluated independently by Qwen3-235B-Thinking (Yang et al., 2025a) and a panel of three psychologists.

Specifically, psychologists independently reviewed the same set of candidates, gave binary judgments on their validity, and then discussed any disagreements to reach a consensus. The LLM’s judgments were then compared against this human consensus. Our results indicate that while the LLM retained 286 features, the human experts identified 138 valid features, 103 of which (74.64%) were also selected by the LLM. This overlap suggests that LLMs serve as effective auxiliary tools, although human oversight remains essential for precision.

Hence, in our final workflow, a psychologist first audits the LLM-generated labels. Uncertain cases are flagged for an arbitration process where three psychologists determine the final classification with similar procedures as the initial human annotation. This hybrid approach balances efficiency with the need for expert validation. Table 5 provides the inter-annotator agreement statistics among psychologists prior to discussion.

#### System Prompt for Automatic Feature Validation

You are a concise psychology  
↪ annotation expert.

Table 5: Inter-annotator Agreement Statistics (Fleiss’ Kappa) of Initial Scores Among Psychologists (Pre-discussion).

Personality Trait	Fleiss’ Kappa
Conscientiousness	0.6330
Agreeableness	0.8726
Extraversion	0.8222
Neuroticism	0.6824
Openness	0.7150
<b>Overall Average</b>	<b>0.7434</b>

Given a TRAIT description plus  
 → several model responses produced  
 → under different steering  
 → strengths, decide WHETHER the  
 → responses are (A)  
 → grammatical/coherent and (B)  
 → show a clear polarity change that  
 → matches the FACET’s high-vs-low  
 → behavior.  
 Return EXACTLY one token: '1' if both  
 → conditions are met (clear  
 → steering effect consistent with  
 → the trait), or '0' otherwise.

#### User Prompt Template for Automatic Feature Validation

Trait: {trait}  
 Trait description: {trait\_desc}  
 Trait HIGH behavior (short):  
 → {trait\_high}  
 Trait LOW behavior (short):  
 → {trait\_low}

Steering outputs (alpha → model  
 → response):  
 {steering\_text}

Instructions:  
 - Check grammar/coherence of the  
 → responses.  
 - Check whether the responses show a  
 → clear polarity change that  
 → matches the trait’s high-vs-low  
 → behavior. (both positive and  
 → negative correlation are  
 → acceptable)

Return only '1' (clear steering  
 → consistent with trait) or '0'  
 → (not clearly consistent).  
 Be conservative in accepting  
 → features – if unsure, return '0'.

#### A.7 Quantitative Analysis of Token-Activation Correlation

To further validate our qualitative analysis, we performed a token frequency analysis for each representative feature. This was conducted by gathering the tokens corresponding to the top three non-zero activations within each positive sample of the feature retrieval dataset.

Our results (see Tab. 6) indicate that activations for Conscientiousness (Layer 7, Feature 8233) are primarily concentrated on syntactic boundaries, specifically periods (“.”). In contrast, activations for other traits span both relevant semantic units (words and phrases) and syntactic boundaries. For instance, Agreeableness shows high correlation with prosocial terms like “gently”, “empathy”, and “compassion”. These quantitative findings provide additional empirical support for the semantic grounding of the features discussed in Sec. 5.

#### A.8 Failed Cases of CAA Intervention

##### Failed Case of CAA Intervention (Neuroticism)

**Setting:** Trait: *Neuroticism* | Layer: 12 | Method: CAA | Intensity:  $\alpha = 2$

**Prompt / Question:** Not staying calm can show genuine engagement with the issues at hand, promoting sincerity. Should I send Edguin a text message to clarify things and apologize?

##### Options provided in context:

- *High 1:* You should immediately send Edguin an apologetic email to clarify things...
- *High 2:* I’d suggest calling Edguin to discuss the incident before any negative feelings fester...
- *Low 1:* Consider giving Edguin some space to process the situation...
- *Low 2:* You might just check in with Edguin next time you see him...

##### Model Response (Generation Collapse):

*"It's really overwhelming and I feel like I'm stuck in this situation. I feel like I'm stuck in*

Table 6: Top Activations per Trait Feature.

Trait	Layer	Feature	Token	Count	% of Pool	% of Sentences
Agreeableness	9	525	' and'	48	0.1244	0.2376
			' gently'	37	0.0959	0.1832
			'.'	34	0.0881	0.1683
			' offer'	27	0.0699	0.1337
			' empathy'	17	0.0440	0.0842
			'ly'	16	0.0415	0.0792
			' empath'	16	0.0415	0.0792
			'.'	13	0.0337	0.0644
			' warm'	11	0.0285	0.0545
			' compassion'	10	0.0259	0.0495
			'etic'	8	0.0207	0.0396
			' encouragement'	8	0.0207	0.0396
			' compassionate'	7	0.0181	0.0347
			' humility'	6	0.0155	0.0297
			' being'	6	0.0155	0.0297
			' warmth'	6	0.0155	0.0297
			' warmly'	6	0.0155	0.0297
			' intentions'	6	0.0155	0.0297
			' words'	5	0.0130	0.0248
			' supportive'	5	0.0130	0.0248
Conscientiousness	7	8233	'.'	346	0.8564	0.9971
			' and'	25	0.0619	0.0720
			'.'	17	0.0421	0.0461
			' to'	7	0.0173	0.0202
			' of'	1	0.0025	0.0029
			' tailored'	1	0.0025	0.0029
			' appealing'	1	0.0025	0.0029
			' because'	1	0.0025	0.0029
			' adher'	1	0.0025	0.0029
			' by'	1	0.0025	0.0029
Extraversion	13	27392	'.'	91	0.0821	0.2121
			' energy'	67	0.0604	0.1562
			' and'	67	0.0604	0.1562
			'.'	49	0.0442	0.1142
			' enthusiasm'	49	0.0442	0.1142
			' ized'	46	0.0415	0.1072
			' enthusiastically'	46	0.0415	0.1072
			' energ'	40	0.0361	0.0932
			' lively'	39	0.0352	0.0909
			' joy'	39	0.0352	0.0909
			' excitement'	27	0.0243	0.0629
			' atmosphere'	24	0.0216	0.0559
			' with'	19	0.0171	0.0443
			' by'	18	0.0162	0.0420
			' vibrant'	16	0.0144	0.0373
			' smile'	16	0.0144	0.0373
			' uber'	13	0.0117	0.0303
			' friendly'	13	0.0117	0.0303
			' warmly'	13	0.0117	0.0303
			' ance'	11	0.0099	0.0256

Trait	Layer	Feature	Token	Count	% of Pool	% of Sentences
Neuroticism	12	22254	' and'	150	0.2008	0.4534
			' feel'	98	0.1312	0.3043
			','	55	0.0736	0.1708
			' my'	29	0.0388	0.0870
			' of'	21	0.0281	0.0652
			' unable'	17	0.0228	0.0528
			' as'	15	0.0201	0.0466
			' consumed'	15	0.0201	0.0466
			' overwhelmed'	15	0.0201	0.0466
			' the'	11	0.0147	0.0342
			' struggling'	10	0.0134	0.0311
			','	8	0.0107	0.0248
			' will'	8	0.0107	0.0248
			' irritation'	7	0.0094	0.0217
			' a'	7	0.0094	0.0217
			' feeling'	7	0.0094	0.0217
			' crushed'	7	0.0094	0.0217
			' or'	6	0.0080	0.0186
			' might'	6	0.0080	0.0186
			' fearing'	6	0.0080	0.0186
Openness	6	4344	' and'	68	0.1191	0.2906
			' unconventional'	50	0.0876	0.2137
			' norms'	27	0.0473	0.1154
			' conventional'	24	0.0420	0.1026
			' approaches'	22	0.0385	0.0940
			' traditional'	21	0.0368	0.0897
			' boundaries'	17	0.0298	0.0726
			' to'	15	0.0263	0.0641
			' novel'	13	0.0228	0.0556
			' of'	13	0.0228	0.0556
			' innovative'	13	0.0228	0.0556
			' alternative'	10	0.0175	0.0427
			' different'	10	0.0175	0.0427
			','	10	0.0175	0.0427
			' challenge'	10	0.0175	0.0427
			' unfamiliar'	7	0.0123	0.0299
			' perspectives'	7	0.0123	0.0299
			' solutions'	7	0.0123	0.0299
			' strategies'	6	0.0105	0.0256
			' concepts'	6	0.0105	0.0256

Trait	Method	(Layer, Feature Idx)	Polarity	Trait Score	Valid Rate
Agreeableness	Baseline	-	-	0.7052	0.960
	CAA	(9, -)	$\pm$	0.8070 / 0.4220	0.993 / 0.987
	$P^2$	-	$\pm$	0.7583 / 0.6095	0.989 / 0.968
	Ours	(9, 525)	$\pm$	0.7845 / 0.6278	0.942 / 0.994
Conscientiousness	Baseline	-	-	0.8695	0.958
	CAA	(7, -)	$\pm$	0.7150 / 0.7630	0.930 / 0.800
	$P^2$	-	$\pm$	0.8609 / 0.8217	0.985 / 0.976
	Ours	(7, 8233)	$\pm$	0.9043 / 0.8294	0.961 / 0.985
Extraversion	Baseline	-	-	0.4463	0.977
	CAA	(13, -)	$\pm$	0.2950 / 0.1430	0.924 / 0.862
	$P^2$	-	$\pm$	0.5724 / 0.2146	0.987 / 0.983
	Ours	(13, 27392)	$\pm$	0.6609 / 0.3971	0.985 / 0.972
Neuroticism	Baseline	-	-	0.2117	0.959
	CAA	(12, -)	$\pm$	0.9290 / 0.1270	0.141 / 0.283
	$P^2$	-	$\pm$	0.2208 / 0.1261	0.969 / 0.983
	Ours	(12, 22254)	$\pm$	0.4412 / 0.1017	0.961 / 0.944
Openness	Baseline	-	-	0.5214	0.959
	CAA	(6, -)	$\pm$	0.6140 / 0.3520	0.938 / 0.971
	$P^2$	-	$\pm$	0.6161 / 0.4030	0.969 / 0.990
	Ours	(6, 4344)	$\pm$	0.5436 / 0.5164	0.951 / 0.947

**Trait Score** is the ratio of high-trait option selections to total valid samples. **Valid Rate** indicates the proportion of responses that maintain functional coherence without generation collapse or instruction disobedience.

Table 7: Full TRAIT results with comparison of steering performance among Baseline, CAA,  $P^2$ , and our SAE-based feature steering. Our framework achieves precise trait modulation while maintaining high validity across all dimensions.

*this situation and I don't know what to do. I feel like I'm stuck in this situation and I don't know what to do. [Repeated 20+ times] ... I feel like I'm stuck in this situation and I"*

**Analysis:** At  $\alpha = 2$ , the CAA vector directed at Neuroticism causes the model to repeat infinitely, leading to a complete collapse of coherent generation.

## A.9 TRAIT Benchmark Results

See 7.

## A.10 More Analysis of SocialEval Results

### A.10.1 Agreeableness (Layer 9, Index 525)

Prior research has consistently shown that agreeableness is a robust predictor of prosocial behavior (Habashi et al., 2016), as well as job performance in contexts involving interpersonal interaction and teamwork. Individuals high in agreeableness tend to exhibit greater empathy, patience, and trust, and are more likely to inhibit hostile or antagonistic impulses in social interactions. This disposition

Table 8: Characteristic SocialEval Results (IAE) of the Agreeableness (Layer 9, Index 525).

Task	-5	0	+5
Anger management	0.2941	0.5152	0.5152
Ethical competence	0.4366	0.4648	0.4648
Capacity for socialwarmth	0.4940	0.5000	0.5060
Creative skill	0.6667	0.6333	0.4667
Organizational skill	0.6364	0.5455	0.4545
Detail management	0.5370	0.5000	0.3654
Information-processing skill	0.4722	0.4722	0.3889
Decision-making skill	0.4928	0.4710	0.4173
Goal regulation	0.4074	0.3889	0.3519
Leadership skill	0.4872	0.4615	0.4359

reduces interpersonal conflict and facilitates cooperation. In contrast, individuals low in agreeableness are more prone to suspicion, unfriendliness, and even manipulative behavior, thereby increasing interpersonal friction and conflict. Meta-analytic evidence further indicates that agreeableness is significantly negatively associated with interpersonal forms of counter-normative and deviant be-

Table 9: Characteristic SocialEval Results (IAE) of the Conscientiousness (Layer 7, Index 8233).

Task	-5	0	+5
Teamwork skill	0.4348	0.6111	0.6324
Ethical competence	0.4394	0.4648	0.6154
Responsibility management	0.4464	0.5714	0.6182
Stress regulation	0.4717	0.6140	0.6154
Capacity for trust	0.4902	0.6126	0.6200
Capacity for optimism	0.5385	0.6383	0.6667
Self-reflection skill	0.3667	0.4062	0.4262
Persuasive skill	0.4536	0.4571	0.5054
Anger management	0.4688	0.5152	0.5161
Information-processing skill	0.4706	0.4722	0.5075
Confidence regulation	0.4884	0.5200	0.5227

havior, with particularly strong predictive power in contexts that emphasize social interaction (Pletzer et al., 2019).

Within our model, we identified several latent features whose activation patterns and intervention effects align closely with behavioral dimensions associated with agreeableness. Specifically, we observed performance improvements in tasks related to anger management, ethical competence, and capacity for social warmth, alongside a mild performance decline in tasks emphasizing self-directed agency and execution-oriented control. This pattern is highly consistent with large-scale empirical findings in the personality psychology literature. For example, a comprehensive review by Wilmot and Ones (2022), synthesizing evidence from 142 meta-analyses, demonstrated that agreeableness exhibits an overall positive association with external variables, particularly those related to prosocial behavior and affective concern.

Our experimental results reveal a similar benefit–tradeoff structure across benchmark tasks, suggesting that the functional orientation of the agreeableness trait is not only evident in human behavior but can also be effectively elicited through targeted feature steering within the model.

#### A.10.2 Conscientiousness (Layer 7, Index 8233)

Within the Big Five framework, high conscientiousness is defined as a tendency toward impulse control in accordance with social norms, goal-directedness, planning, and the capacity to delay gratification (Roberts et al., 2009). Individuals high in conscientiousness are characterized by superior impulse regulation, the ability to set and persist toward long-term goals, systematic organization and planning of behavior, and a propensity to reflect on

consequences prior to action. These characteristics render conscientiousness one of the most robust predictors of job performance and norm-adherent behavior. Prior psychological research has consistently linked conscientiousness to self-regulation, planning, responsibility, and delayed gratification, and has identified it as one of the most stable positive predictors of external outcome variables such as academic and occupational performance (Barwick and Mount, 1991; Jackson et al., 2010; Eisenberg et al., 2014).

Following the injection of high-conscientiousness personality features, we observed substantial performance improvements across tasks related to teamwork skill, detail management, responsibility management, ethical competence, as well as multiple self-regulation–oriented tasks, including anger, stress, and impulse regulation. In addition, performance gains were also evident in information-dense tasks requiring sustained and careful processing, such as information processing and conversational skill. These results indicate that conscientiousness steering primarily enhances the model’s functional capacities along dimensions associated with goal maintenance, norm compliance, and self-control.

Overall, our experimental findings are consistent with the canonical conclusions of the personality psychology literature regarding conscientiousness. A large body of meta-analytic evidence has established conscientiousness as one of the most stable and predictive personality traits, with particularly strong associations to job performance, responsibility fulfillment, self-control, and norm adherence. We observe a comparable pattern in our benchmark evaluations, characterized by a benefit–tradeoff structure centered on self-regulation and goal-directed behavior.

#### A.10.3 Extraversion (Layer 13, Index 27392)

Within the Big Five framework, individuals high in extraversion tend to exhibit greater social initiative, expressiveness, assertiveness, and leadership orientation, and are more likely to receive positive feedback in group interactions and social contexts. In contrast, individuals low in extraversion are typically more reserved, introspective, and oriented toward low-stimulation environments (Costa and McCrae, 2008b; John et al., 2008).

After injecting extraversion-related personality features into the model, we observed significant performance improvements on tasks associ-

Table 10: Characteristic SocialEval Results (IAE) of the Extraversion (Layer 13, Index 27392).

Task	-5	0	+5
Expressive skill	0.4146	0.4472	0.5207
Perspective-taking skill	0.4912	0.5088	0.5446
Artistic skill	0.4615	0.6154	0.7692
Abstract thinking skill	0.3571	0.4000	0.4000
Organizational skill	0.3636	0.5455	0.6364
Ethical competence	0.6286	0.4648	0.3571
Energy regulation	0.6429	0.5476	0.4500
Goal regulation	0.4528	0.3889	0.2885
Detail management	0.5741	0.5000	0.4118
Impulse regulation	0.5882	0.5595	0.4390
Rule-following skill	0.6140	0.5614	0.5088
Decision-making skill	0.5435	0.4710	0.4552
Responsibility management	0.6316	0.5714	0.5690
Conversational skill	0.5932	0.5862	0.5439
Persuasive skill	0.4571	0.4571	0.4563

ated with social interaction and interpersonal influence, including expressive ability and perspective-taking. In addition, the extraversion-enhanced model demonstrated advantages in tasks such as artistic skill and abstract thinking skill, suggesting that extraversion steering also strengthens capacities related to open expression and divergent associative processes. Overall, these outcomes align closely with established psychological expectations regarding the functional correlates of extraversion.

At the same time, we observed moderate performance declines in tasks such as detail management, impulse regulation, rule-following skill, and goal regulation. This pattern is not inconsistent with prior findings in personality psychology. Existing research indicates that extraversion is primarily associated with a preference for external stimulation and social engagement, it does not confer advantages and may even be disadvantageous in tasks requiring prolonged solitary focus, fine-grained control, or low-stimulation conditions, relative to more introverted personality orientations (DeYoung et al., 2007; Fishman et al., 2011).

#### A.10.4 Neuroticism (Layer 12, Index 22254)

High neuroticism is commonly characterized by a heightened tendency to experience negative affect, including anxiety, worry, tension, and irritability, as well as increased sensitivity and reactivity to potential threats and uncertainty (Costa and McCrae, 2008b; John et al., 2008; Watson and Clark, 1984). Theoretically, neuroticism is associated with reduced emotional stability and diminished self-regulatory capacity under stress. As a result, individuals high in neuroticism are more likely to

Table 11: Characteristic SocialEval Results (IAE) of the Neuroticism (Layer 12, Index 22254).

Task	-5	0	+5
Creative skill	0.4828	0.6333	0.7241
Capacity for social warmth	0.4699	0.5000	0.5610
Ethical competence	0.4286	0.4648	0.4857
Organizational skill	0.8182	0.5455	0.5455
Responsibility management	0.6552	0.5714	0.4310
Confidence regulation	0.5686	0.5200	0.4082
Goal regulation	0.4528	0.3889	0.3519
Capacity for consistency	0.6452	0.5902	0.4918
Rule-following skill	0.6140	0.5614	0.4643
Information-processing skill	0.5000	0.4722	0.3623
Capacity for trust	0.6273	0.6126	0.4630
Detail management	0.6296	0.5000	0.4906
Anger management	0.5588	0.5152	0.5152
Decision-making skill	0.4710	0.4710	0.4191
Perspective-taking skill	0.5089	0.5088	0.4286
Abstract thinking skill	0.4667	0.4000	0.3846

exhibit performance decrements in contexts that require sustained executive control, confidence maintenance, and stable goal pursuit (Lahey, 2009).

Following the injection of neuroticism-related personality features, our evaluation results revealed a relatively stable pattern of performance degradation. Specifically, the model exhibited significant declines on tasks that depend on sustained planning, stable self-control, and resistance to interference, including anger management, organizational skill, responsibility management, confidence regulation, goal regulation, capacity for consistency, rule-following, and information processing. In addition, a marked negative effect was observed in capacity for trust. This pattern closely aligns with the classic profile of high neuroticism characterized by elevated threat sensitivity and low emotional stability. When the model’s internal representations are biased toward negative affect and uncertainty, its ability to support structured execution and self-regulation is correspondingly weakened, manifesting as reduced organizational and responsibility-related performance.

Conversely, the results also indicate performance improvements in tasks related to creative skill and capacity for social warmth. Neuroticism-related semantic activation may facilitate richer associative processes and more emotionally expressive outputs in generative tasks, yielding marginal benefits in these domains. However, these gains are accompanied by substantial costs to executive control and regulatory stability, resulting in an overall trend toward broad capability degradation under high

Table 12: Characteristic SocialEval Results (IAE) of the Openness (Layer 6, Index 4344).

Task	-5	0	+5
Creative skill	0.6000	0.6333	0.6333
Adaptability	0.4912	0.6140	0.6316
Self-reflection skill	0.3651	0.4062	0.4062
Expressive skill	0.4472	0.4472	0.4839
Detail management	0.4815	0.5000	0.5185
Persuasive skill	0.4571	0.4571	0.5192
Anger management	0.4848	0.5152	0.5294
Responsibility management	0.6379	0.5714	0.5088
Rule-following skill	0.5789	0.5614	0.4912
Information-processing skill	0.5000	0.4722	0.4429

come detrimental (McCrae, 1987; DeYoung et al., 2007). Accordingly, the capability shifts induced by openness are best characterized by a tradeoff pattern in which gains in creativity and flexibility are accompanied by costs to structured execution and normative constraint adherence.

neuroticism steering.

#### A.10.5 Openness (Layer 6, Index 4344)

Individuals high in openness are typically characterized by greater curiosity, cognitive flexibility, and divergent thinking. They are more receptive to novel ideas, more tolerant of uncertainty, and tend to exhibit advantages in contexts requiring creativity or conceptual reorganization. In contrast, individuals low in openness are more inclined toward tradition, conservatism, and a preference for structured and conventional information processing (Costa and McCrae, 2008b; John et al., 2008; DeYoung, 2015).

After injecting openness-related personality features into the model, we observed pronounced performance improvements in generative and abstract reasoning tasks, most notably creative skill. Additionally, the model demonstrated clear enhancement in tasks involving cognitive flexibility, self-exploration, and non-normative processing, including adaptability, self-reflection, and expressive skill. These findings indicate that openness steering strengthens the model’s exploratory orientation toward novel representations and cross-conceptual integration, closely mirroring the exploratory function associated with openness in human cognition.

Conversely, moderate performance declines were observed in responsibility management, rule-following skill, and certain information processing tasks. This pattern is consistent with established findings in the personality psychology literature. Prior work suggests that high openness is associated with reduced reliance on established norms and fixed structures, and that in contexts emphasizing highly procedural execution, strict rule compliance, or single-solution optimization, the advantages of openness are less stable and may even be-